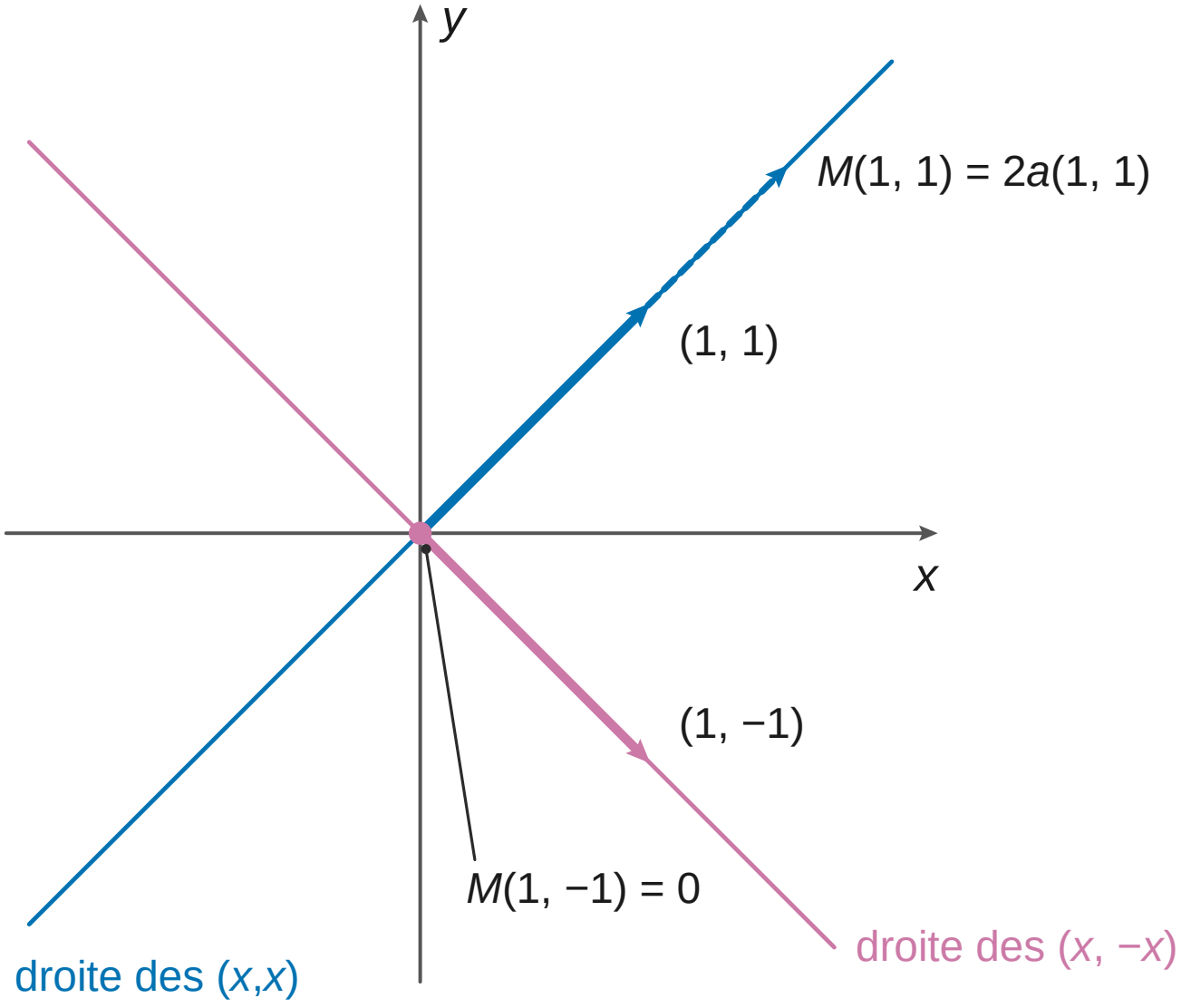

$n = 1, A = (a)$

$$P = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}$$

$$D = P^{-1}MP$$

$$D = \begin{pmatrix} 0 & 0 \\ 0 & 2a \end{pmatrix}$$